

Differentiation

4.1 Derivatives

Consider the curve shown Figure 4.1. It is clear from intuition that the “*slope*” changes as we move along the curve. At P' , the slope is very steep whereas at P , the slope is gentle (in this sentence, slope means a piece of ground going up or down).

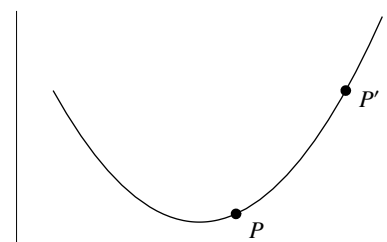


Figure 4.1

In elementary coordinate geometry, readers have learnt the concept “*slope of a line*”. It is a number which measures how steep is the line. For a non-vertical line, its slope is given by

$$\frac{y_2 - y_1}{x_2 - x_1}$$

where (x_1, y_1) and (x_2, y_2) are two distinct points on the line and the value is independent of the choice of the two points.

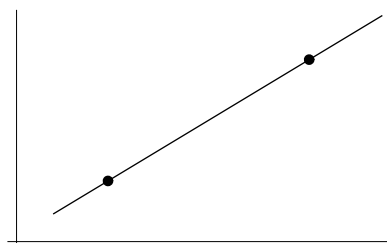


Figure 4.2

For curves, we shouldn’t say “*slope of a curve*” because at different points of the curve, the slopes are different. Instead we should say “*slope of a curve at a point*”. Below is how we define this concept.

First we have a curve $\mathcal{C}$ and a point P on the curve. To define the slope of $\mathcal{C}$ at P , take a point Q on the curve different from P . The line PQ is called a *secant line at P*. Its slope, denoted by m_{PQ} , can be found using the coordinates of P and Q . If we let Q move along the curve, the slope m_{PQ} changes.

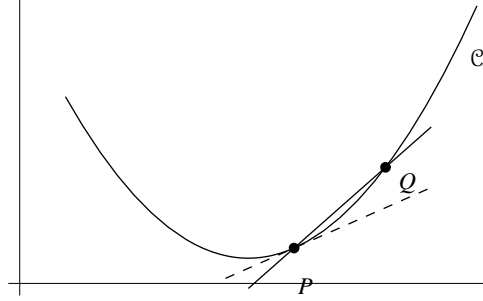


Figure 4.3

Suppose that as Q approaches P , the number m_{PQ} approaches a fixed value. This value, denoted by $m_{\mathcal{C},P}$ or simply m_P if the curve is understood, is called the *slope of $\mathcal{C}$ at P* ; and the line with slope m_P and passing through P is called the *tangent line to the curve $\mathcal{C}$ at P* .

Remark The number $m_{\mathcal{C},P}$ (if exist) is the unique real number satisfying

(*) m_{PQ} is arbitrarily close to $m_{\mathcal{C},P}$ if Q belonging to $\mathcal{C}$ is sufficiently close to (but different from) P .

In view of the concept “*limit of a function at a point*” and the notation $\lim_{x \rightarrow a} f(x)$, we may write

$$\lim_{\substack{Q \rightarrow P \\ \text{along } \mathcal{C}}} m_{PQ} = m_{\mathcal{C},P}$$

to mean that (*) holds. Below, we will discuss how to find $\lim_{\substack{Q \rightarrow P \\ \text{along } \mathcal{C}}} m_{PQ}$ by rewriting it as the limit of a difference quotient.

Formula for Slope Suppose $\mathcal{C}$ is given by $y = f(x)$, where f is a function; and $P(x_0, f(x_0))$ is a point on $\mathcal{C}$. For any point Q on $\mathcal{C}$ with $Q \neq P$, its x -coordinate can be written as $x_0 + h$ where $h \neq 0$ (if $h > 0$, Q is on the right of P ; if $h < 0$, Q is on the left of P). Thus, Q can be written as $(x_0 + h, f(x_0 + h))$. The slope m_{PQ} of the secant line PQ is

$$\begin{aligned} m_{PQ} &= \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0} \\ &= \frac{f(x_0 + h) - f(x_0)}{h} \end{aligned}$$

Note that as Q approaches P , the number h approaches 0. From these, we see that the slope of $\mathcal{C}$ at P (denoted by m_P) is

$$m_P = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (4.1.1)$$

provided that the limit exists.

Remark The limit in (4.1.1) is a two-sided limit. This is because Q can approach P from the left or from the right and so h can approach 0 from the left or from the right.

4.1. Derivatives

Example Find the slope of the curve given by $y = x^2$ at the point $P(3, 9)$.

Solution Put $f(x) = x^2$. By (4.1.1), the required slope (denoted by m_P) is

$$\begin{aligned}
 m_P &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(3+h)^2 - 3^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(9 + 6h + h^2) - 9}{h} \\
 &= \lim_{h \rightarrow 0} \frac{6h + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (6 + h) \\
 &= 6.
 \end{aligned}$$

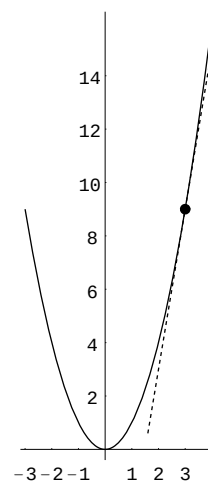


Figure 4.4

Definition Let x_0 be a real number and let f be a function defined on an open interval containing x_0 . Suppose the limit in (4.1.1) exists. Then we say that f is *differentiable* at x_0 .

Convention Open intervals will be denoted by (a, b) including the cases where $a = -\infty$ and/or $b = \infty$, unless otherwise stated. Thus (a, b) can be any one of the following:

- (a, b) where $a, b \in \mathbb{R}$ and $a < b$;
- $(-\infty, b)$ where $a = -\infty$ and $b \in \mathbb{R}$;
- (a, ∞) where $a \in \mathbb{R}$ and $b = \infty$;
- $(-\infty, \infty)$ where $a = -\infty$ and $b = \infty$.

Remark

- The condition “ f is a function defined on an open interval containing x_0 ” means that there is an open interval (a, b) such that $(a, b) \subseteq \text{dom}(f)$ and $x_0 \in (a, b)$. Hence, f is defined on the left-side and right-side of x_0 as well as at x_0 . The expression $\frac{f(x_0+h) - f(x_0)}{h}$ in the limit in (4.1.1), considered as a function of h , is defined on the left-side and the right-side of 0 but is undefined at 0.
- “ f is differentiable at x_0 ” means that the slope of the curve $\mathcal{C}$ at P exists, where $\mathcal{C}$ is given by $y = f(x)$ and P is the point on $\mathcal{C}$ whose x -coordinate is x_0 .
- There is an alternative way to describe the limit in (4.1.1). Putting $x = x_0 + h$, we have $x - x_0 = h$. Note that as h approaches to 0, x approaches to x_0 . Hence we have

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Theorem 4.1.1 Let x_0 be a real number and let f be a function defined on an open interval containing x_0 . Suppose f is differentiable at x_0 . Then f is continuous at x_0 .

Proof Since f is differentiable at x_0 , by definition, $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ exists (a real number). Hence we have

$$\begin{aligned}
 \lim_{x \rightarrow x_0} (f(x) - f(x_0)) &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) \right) \\
 &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \right) \cdot \lim_{x \rightarrow x_0} (x - x_0) && \text{Limit Rule (La5)} \\
 &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \right) \cdot (x_0 - x_0) && \text{Theorem 3.5.1} \\
 &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \right) \cdot 0 \\
 &= 0.
 \end{aligned}$$

Therefore, we get

$$\begin{aligned}
 \lim_{x \rightarrow x_0} f(x) &= \lim_{x \rightarrow x_0} (f(x) - f(x_0) + f(x_0)) \\
 &= \lim_{x \rightarrow x_0} (f(x) - f(x_0)) + \lim_{x \rightarrow x_0} f(x_0) && \text{Limit Rule (La4)} \\
 &= 0 + f(x_0) && \text{From above and Limit Rule (La1)} \\
 &= f(x_0).
 \end{aligned}$$

That is, f is continuous at x_0 .

The following example illustrates that converse of Theorem 4.1.1 is not true.

Example Let $f(x) = |x|$. The domain of f is $\mathbb{R}$.

The function f is continuous at 0. This is because

- $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-x) = 0$;
- $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0$,

and so $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$.

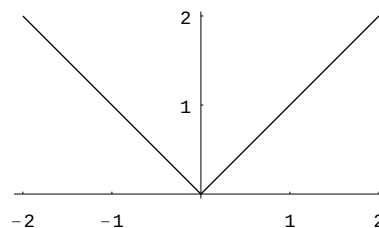


Figure 4.5

However, f is not differentiable at 0. This is because $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$ does not exist as the left-side and right-side limits are unequal:

$$\begin{aligned}
 \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{h - 0}{h} && \text{and} && \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^-} \frac{-h - 0}{h} \\
 &= \lim_{h \rightarrow 0^+} 1 && && &= \lim_{h \rightarrow 0^-} -1 \\
 &= 1 && && &= -1.
 \end{aligned}$$

Definition Let f be a function.

- (1) Suppose that f is differentiable at every point belonging to an open interval (a, b) . Then we say that f is *differentiable on (a, b)* .

- (2) Suppose that f is differentiable at every point belonging to its domain. Then we say that f is a *differentiable function*.

Remark If f is a differentiable function, then its domain can be written as a union of open intervals.

Example Let $f(x) = |x|$. In a previous example, we have seen that f is not differentiable at 0. Thus f is not a differentiable function. Below we will show that f is differentiable at every $x_0 \neq 0$. Thus, f is differentiable on $(0, \infty)$ as well as on $(-\infty, 0)$.

($x_0 > 0$) In this case, if h is a small enough real number, then $x_0 + h > 0$ and so we have

$$\begin{aligned} f(x_0 + h) - f(x_0) &= |x_0 + h| - |x_0| \\ &= (x_0 + h) - x_0 = h, \end{aligned}$$

which yields

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{h}{h} \\ &= \lim_{h \rightarrow 0} 1 = 1. \end{aligned}$$

($x_0 < 0$) In this case, if h is a small enough real number, then $x_0 + h < 0$ and so we have

$$\begin{aligned} f(x_0 + h) - f(x_0) &= |x_0 + h| - |x_0| \\ &= -(x_0 + h) - (-x_0) = -h, \end{aligned}$$

which yields

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{-h}{h} \\ &= \lim_{h \rightarrow 0} -1 = -1. \end{aligned}$$

A function f is differentiable means that for every $x \in \text{dom}(f)$, the limit of the difference quotient $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists; the limit is a real number and its value depends on x . In this way, we get a function, called the *derivative* of f and denoted by f' , from $\text{dom}(f)$ to $\mathbb{R}$.

The graph of f is a curve. The assumption that f is a differentiable function implies that at every point on the curve, the slope exists; at the point whose x -coordinate is x_0 , the slope is $f'(x_0)$. Thus f' can be considered as *slope function*.

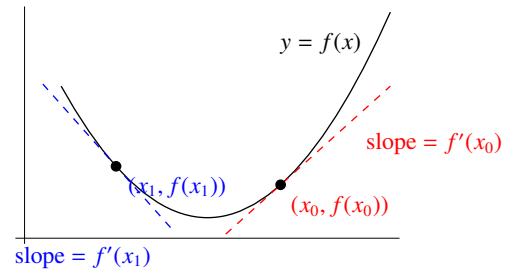


Figure 4.6

More generally, if f is differentiable only at some points in its domain, we can still define the derivative of f on a smaller set.

Definition Let f be a function that is differentiable at some points belonging in its domain. Then the *derivative* of f , denoted by f' , is the function (from a subset of $\text{dom}(f)$ into $\mathbb{R}$) given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

where the domain of f' is $\left\{x \in \text{dom}(f) : \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ exists} \right\}$.

Example Let $f(x) = |x|$. Using results in a previous example, we see that

$$f'(x) = \begin{cases} 1 & \text{if } x > 0, \\ -1 & \text{if } x < 0, \end{cases}$$

where $\text{dom}(f') = \{x \in \mathbb{R} : x \neq 0\}$.

Example Let $f(x) = x^2$. Find the derivative of f .

Explanation To find the derivative of f means to find the domain of f' and find a formula for $f'(x)$.

Solution By definition, we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) \\ &= 2x. \end{aligned}$$

The domain of f' is $\mathbb{R}$.

Remark The logic in the above solution is as follows:

- (1) First we find that $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 2x$ for all $x \in \mathbb{R}$.
- (2) From (1), we see that the domain of f' is $\mathbb{R}$ and $f'(x) = 2x$ for all $x \in \mathbb{R}$.

Example Let $f(x) = x^3$. Find $f'(x)$.

Solution By definition, we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x^3 + 3x^2h + 3xh^2 + h^3) - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) \\ &= 3x^2. \end{aligned}$$

4.1. Derivatives

Remark The domain of f' is $\mathbb{R}$.

Terminology The process of finding derivatives is called *differentiation*.

In a previous example to find the slope of the parabola $y = f(x) = x^2$ at the point $(3, 9)$, we use definition to find $f'(3)$. In fact, if we know that $f'(x) = 2x$, then by direct substitution, we get $f'(3) = 6$. In the next section, we will discuss how to find $f'(x)$ using rules for differentiation.

Terminology $f'(x_0)$ is called the *derivative of f at x_0* .

To represent a function f , we sometimes write y or $f(x)$. Similarly, the derivative of a function can be represented in many ways.

Notation

- To denote the derivative of a function f , we have the following notations.

$$f', \quad y', \quad \frac{dy}{dx}, \quad Df, \quad Dy, \quad f'(x) \quad \text{and} \quad \frac{d}{dx}f(x).$$

- To denote the derivative of f at x_0 , we have the following notations.

$$f'(x_0), \quad y'(x_0), \quad \left. \frac{dy}{dx} \right|_{x=x_0}, \quad Df(x_0) \quad \text{and} \quad Dy(x_0).$$

Some readers may wonder why we have the $'$ notation as well as the $\frac{d}{dx}$ notation. Calculus was “*invented*” by Newton and Leibniz independently in the late 17th century. Newton used $\dot{x}$ whereas Leibniz used $\frac{dx}{dt}$ to denote the derivative of x with respect to t (time). The notation y' is simple whereas $\frac{dy}{dx}$ reminds us that it is defined as a limit of *difference quotient*:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

where $\Delta x = h = (x + h) - x$ and $\Delta y = f(x + h) - f(x)$ are changes in x and y respectively.

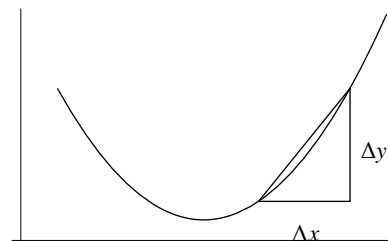


Figure 4.7

Caution $\frac{dy}{dx}$ is not a fraction.

- $\frac{dy}{dx}$ is f' or $f'(x)$; it is a function or an expression in x . It can be written as $\frac{d}{dx}y$ also. The notation $\frac{d}{dx}$ can be considered as an operation; to find $\frac{dy}{dx}$ means to perform the differentiation operation on y . Some authors use the notation Dy instead, where D stands for the differentiation operator (some authors use $D_x y$ to emphasize that the variable is x).
- Although we can define dy and dx (called *differentials*), $\frac{dy}{dx}$ does not mean “divide dy by dx ”. In Chapter 10, we will describe *differentials* briefly (the purpose is to introduce the substitution method for integration).

Using $\frac{d}{dx}$ notation, the results obtained in the last two examples can be written as

$$\frac{d}{dx}x^2 = 2x$$

$$\frac{d}{dx}x^3 = 3x^2.$$

In Exercise 3.5, Question 2(c) and (d), in terms of the $\frac{d}{dx}$ notation, the answers can be written as

$$\frac{d}{dx}x^{-1} = -x^{-2}$$

$$\frac{d}{dx}x^{\frac{1}{2}} = \frac{1}{2}x^{-\frac{1}{2}}.$$

These are particular cases of a general result, called the *power rule* which will be discussed in Section 4.2.

Rate of Change

- The slope of a line is the rate of change of y with respect to x . The slope of a curve $y = f(x)$ at a point $P(x_0, f(x_0))$ is the limit of the slopes of the secant lines through P , so it is the rate of change of y with respect to x at P . That is, $f'(x_0)$ or $\left.\frac{dy}{dx}\right|_{x=x_0}$ is the rate of change of y with respect to x when $x = x_0$.
- If $x = t$ is time and $y = s(t)$ is the displacement function of a moving object then $s'(t_0)$ or $\left.\frac{ds}{dt}\right|_{t=t_0}$ is the rate of change of displacement with respect to time when $t = t_0$, that is, the (instantaneous) velocity at $t = t_0$. In the Introduction of Chapter 3, we consider the velocity of an object at time $t = 2$, where the displacement function is $s(t) = t^2$. Using differentiation, the velocity at $t = 2$ can be found easily:

$$\begin{aligned} s'(2) &= \left.\frac{d}{dt}t^2\right|_{t=2} \\ &= 2t\big|_{t=2} \\ &= 4. \end{aligned}$$

Remark The notation $2t\big|_{t=2}$ means substitute $t = 2$ into the expression $2t$. More generally, the notation $f(x)\big|_{x=x_0}$ means $f(x_0)$.

Exercise 4.1

- For each of the following f , use definition to find $f'(x)$.

(a) $f(x) = 2x^2 + 1$	(b) $f(x) = x^3 - 3x$
(c) $f(x) = x^4$	(d) $f(x) = \frac{1}{x^2}$

4.2 Rules for Differentiation

Derivative of Constant The derivative of a constant function is 0 (the zero function), that is

$$\frac{d}{dx}c = 0,$$

where c is a constant.

Explanation In the above formula, we use c to denote the constant function f given by $f(x) = c$. The domain of f is $\mathbb{R}$. The result means that f is differentiable on $\mathbb{R}$ and that $f'(x) = 0$ for all $x \in \mathbb{R}$.

Proof Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = c$ for $x \in \mathbb{R}$. By definition, we get

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} \\ &= \lim_{h \rightarrow 0} \frac{0}{h} \\ &= \lim_{h \rightarrow 0} 0 \\ &= 0 \end{aligned}$$

Geometric meaning The graph of the constant function c is the horizontal line given by $y = c$. At every point on the line, the slope is 0.

Derivative of Identity Function *The derivative of the identity function is the constant function 1, that is*

$$\frac{d}{dx}x = 1.$$

Explanation In the above formula, we use x to denote the *identity function*, that is, the function f given by $f(x) = x$. The domain of f is $\mathbb{R}$. The result means that f is differentiable on $\mathbb{R}$ and $f'(x) = 1$ for all $x \in \mathbb{R}$.

Proof Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x$ for $x \in \mathbb{R}$. By definition, we get

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x+h-x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} \\ &= \lim_{h \rightarrow 0} 1 \\ &= 1 \end{aligned}$$

Geometric meaning The graph of the identity function is the line given by $y = x$. At every point on the line, the slope is 1.

Power Rule for Differentiation (positive integer version) *Let n be a positive integer. Then the power function x^n is differentiable on $\mathbb{R}$ and we have*

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Explanation In the above formula, we use x^n to denote the n -th *power function*, that is, the function f given by $f(x) = x^n$. The domain of f is $\mathbb{R}$. The result means that f is differentiable on $\mathbb{R}$ and $f'(x) = nx^{n-1}$ for all

$x \in \mathbb{R}$. When $n = 1$, the formula becomes $\frac{d}{dx}x = 1x^0$. In the expression on the right side, x^0 is understood to be the constant function 1 and so the formula reduces to $\frac{d}{dx}x = 1$ which is the rule for derivative of the identity function. To prove that the result is true for all positive integers n , we can use mathematical induction. For base step, we know that the result is true when $n = 1$. For the induction step, we can apply product rule (will be discussed later). Below we will give alternative proofs for the power rule (positive integer version).

Proof Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^n$ for $x \in \mathbb{R}$. By definition, we get

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}. \end{aligned}$$

To find the limit, we “simplify” the numerator to obtain a factor h and then cancel it with the factor h in the denominator. For this, we can use:

- a factorization formula $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1})$
- the Binomial Theorem $(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}a^2b^{n-2} + \binom{n}{n-1}ab^{n-1} + b^n$

where $\binom{n}{1} = n$ and $\binom{n}{2} = \frac{n(n-1)}{2}$ and $\binom{n}{3} = \frac{n(n-1)(n-2)}{3 \cdot 2}$ etc. are the binomial coefficients (see Binomial Theorem in the appendix). However, in the proof below, we only need to know the coefficient $\binom{n}{1} = n$; the other coefficients are not important.

(Method 1) By the factorization formula, we get

$$\begin{aligned} (x+h)^n - x^n &= (x+h-x)\left((x+h)^{n-1} + (x+h)^{n-2}x + \cdots + (x+h)x^{n-2} + x^{n-1}\right) \\ &= h\left((x+h)^{n-1} + (x+h)^{n-2}x + \cdots + (x+h)x^{n-2} + x^{n-1}\right) \end{aligned}$$

$$\begin{aligned} \text{Therefore, } f'(x) &= \lim_{h \rightarrow 0} \left((x+h)^{n-1} + (x+h)^{n-2}x + \cdots + (x+h)x^{n-2} + x^{n-1}\right) \\ &= (x+0)^{n-1} + (x+0)^{n-2} \cdot x + \cdots + (x+0) \cdot x^{n-2} + x^{n-1} && \text{Theorem 3.5.1} \\ &= \underbrace{x^{n-1} + x^{n-1} + \cdots + x^{n-1} + x^{n-1}}_{n \text{ terms}} \\ &= nx^{n-1} \end{aligned}$$

(Method 2) From the Binomial Theorem, we get

$$\begin{aligned} (x+h)^n - x^n &= x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-2}x^2h^{n-2} + \binom{n}{n-1}xh^{n-1} + h^n - x^n \\ &= \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + \binom{n}{n-2}x^2h^{n-2} + \binom{n}{n-1}xh^{n-1} + h^n \\ &= h\left(nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + \binom{n}{n-2}x^2h^{n-3} + \binom{n}{n-1}xh^{n-2} + h^{n-1}\right) && \text{because } \binom{n}{1} = n \end{aligned}$$

$$\begin{aligned} \text{Therefore, } f'(x) &= \lim_{h \rightarrow 0} \left(nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + \binom{n}{n-2}x^2h^{n-3} + \binom{n}{n-1}xh^{n-2} + h^{n-1}\right) \\ &= nx^{n-1} + \binom{n}{2}x^{n-2} \cdot 0 + \cdots + \binom{n}{n-2}x^2 \cdot 0 + \binom{n}{n-1}x \cdot 0 + 0 && \text{Theorem 3.5.1} \\ &= nx^{n-1}. \end{aligned}$$

4.2. Rules for Differentiation

Example Let $y = x^{123}$. Find $\frac{dy}{dx}$.

Explanation The notation $y = x^{123}$ represents a power function. To find $\frac{dy}{dx}$ means to find the derivative of the function.

$$\begin{aligned} \text{Solution } \frac{dy}{dx} &= \frac{d}{dx} x^{123} \\ &= 123x^{123-1} && \text{Power Rule} \\ &= 123x^{122} \end{aligned}$$

□

Caution In the above solution, the first step is “*substitution*”. It can also be written as $\frac{dx^{123}}{dx}$. It is wrong to write $\frac{dy}{dx} x^{123}$ which means $\frac{dy}{dx}$ multiplied by x^{123} . The notation $\frac{d}{dx} x^{123}$ is the derivative of the power function x^{123} . If we consider $\frac{d}{dx}$ as an operator, $\frac{d}{dx} x^{123}$ means “*perform the differentiation operation on the function x^{123}* ”.

Remark

- In the power rule, if we put $n = 0$, the left side is $\frac{d}{dx} x^0$ and right side is $0x^{-1}$. The function x^0 is understood to be the constant function 1. Thus $\frac{d}{dx} x^0 = 0$ by the rule for derivative of constant. The domain of the function x^{-1} is $\mathbb{R} \setminus \{0\}$. Therefore, $0x^{-1}$ is the function whose domain is $\mathbb{R} \setminus \{0\}$ and is always equal to 0. If we extend the domain to $\mathbb{R}$ and treat $0x^{-1}$ as the constant function 0, then the power rule is true when $n = 0$.
- Later in this section, we will show that the power rule is also true for negative integers n using the *quotient rule* (in fact, it is true for all real numbers n ; the result will be called the *general power rule*).
- In Chapter 6, we will discuss integration which is the reverse process of differentiation. There is a result called the *power rule for integration*. However, in this chapter, “*power rule*” always means “*power rule for differentiation*”.

Constant Multiple Rule for Differentiation Let f be a function and let k be a constant. Suppose that f is differentiable. Then the function kf is also differentiable. Moreover, we have

$$\frac{d}{dx}(kf)(x) = k \cdot \frac{d}{dx} f(x).$$

Explanation The function kf is defined by $(kf)(x) = k \cdot f(x)$ for $x \in \text{dom}(f)$. The result means that if $f'(x)$ exists for all $x \in \text{dom}(f)$, then $(kf)'(x) = k \cdot f'(x)$ for all $x \in \text{dom}(f)$, that is, $(kf)' = k \cdot f'$.

Proof By definition, we have

$$\begin{aligned} (kf)'(x) &= \lim_{h \rightarrow 0} \frac{(kf)(x+h) - (kf)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{k \cdot f(x+h) - k \cdot f(x)}{h} && \text{Definition of } kf \\ &= \lim_{h \rightarrow 0} \left(k \cdot \frac{f(x+h) - f(x)}{h} \right) \\ &= k \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} && \text{Limit Rule (La5s)} \\ &= k \cdot f'(x) \end{aligned}$$

Remark There is a *pointwise version* of the *constant multiple rule*: the result remains valid if *differentiable function* is replaced by *differentiable at a point*.

Let f be a function and let k be a constant. Suppose that f is differentiable at x_0 . Then the function kf is also differentiable at x_0 . Moreover, we have $(kf)'(x_0) = k \cdot f'(x_0)$

There are also pointwise versions for the *sum rule*, *product rule* and *quotient rule* which will be discussed later. Readers can formulate the results themselves.

Example Let $y = 3x^4$. Find $\frac{dy}{dx}$.

$$\begin{aligned} \text{Solution } \frac{dy}{dx} &= \frac{d}{dx} 3x^4 \\ &= 3 \cdot \frac{d}{dx} x^4 && \text{Constant Multiple Rule} \\ &= 3 \cdot (4x^{4-1}) && \text{Power Rule} \\ &= 12x^3 \end{aligned}$$

Sum Rule for Differentiation (Term by Term Differentiation) Let f and g be functions with the same domain. Suppose that f and g are differentiable. Then the function $f + g$ is also differentiable. Moreover, we have

$$\frac{d}{dx}(f + g)(x) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x).$$

Explanation The function $f + g$ is defined by $(f + g)(x) = f(x) + g(x)$ for x belonging to the common domain A of f and g . The result means that if $f'(x)$ and $g'(x)$ exist for all $x \in A$, then $(f + g)'(x) = f'(x) + g'(x)$ for all $x \in A$, that is, $(f + g)' = f' + g'$.

Proof By definition, we have

$$\begin{aligned} (f + g)'(x) &= \lim_{h \rightarrow 0} \frac{(f + g)(x + h) - (f + g)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(f(x + h) + g(x + h)) - (f(x) + g(x))}{h} && \text{Definition of } f + g \\ &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x) + g(x + h) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{f(x + h) - f(x)}{h} + \frac{g(x + h) - g(x)}{h} \right) \\ &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x + h) - g(x)}{h} && \text{Limit Rule (La4)} \\ &= f'(x) + g'(x) \end{aligned}$$

Remark

- If the domains of f and g are not the same but their intersection is nonempty, we define $f + g$ to be the function with domain $\text{dom}(f) \cap \text{dom}(g)$ given by $(f + g)(x) = f(x) + g(x)$. The following is a more general version of the *sum rule*:

Let f and g be functions. Suppose that f and g are differentiable on an open interval (a, b) . Then the function $f + g$ is also differentiable on (a, b) . Moreover, on the interval (a, b) , we have

$$\frac{d}{dx}(f + g)(x) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x).$$

There are also more general versions for the *product rule* and *quotient rule*. Readers can formulate the results themselves.

- The result is also true for difference of two functions, that is,

$$\frac{d}{dx}(f - g)(x) = \frac{d}{dx}f(x) - \frac{d}{dx}g(x).$$

The result for difference can be proved similar to that for sum. Alternatively, it can be proved from the sum rule together with the constant multiple rule:

$$\begin{aligned}(f - g)'(x) &= (f + (-1)g)'(x) \\ &= f'(x) + ((-1)g)'(x) && \text{Sum Rule} \\ &= f'(x) + (-1) \cdot g'(x) && \text{Constant Multiple Rule} \\ &= f'(x) - g'(x).\end{aligned}$$

- Term by Term Differentiation can be applied to sum and difference of finitely many terms.

Example Let $y = x^2 + 3$. Find $\frac{dy}{dx}$.

$$\begin{aligned}\text{Solution } \frac{dy}{dx} &= \frac{d}{dx}(x^2 + 3) \\ &= \frac{d}{dx}x^2 + \frac{d}{dx}3 && \text{Term by Term Differentiation} \\ &= 2x + 0 && \text{Power Rule and Derivative of Constant} \\ &= 2x\end{aligned}$$

Example Let $f(x) = x^5 - 6x^7$. Find $f'(x)$.

Explanation This question is similar to the last one. If we put $y = f(x)$, then $f'(x) = \frac{dy}{dx}$. Below, we use the notation $\frac{d}{dx}$ to perform differentiation.

$$\begin{aligned}\text{Solution } f'(x) &= \frac{d}{dx}(x^5 - 6x^7) \\ &= \frac{d}{dx}x^5 - \frac{d}{dx}6x^7 && \text{Term by Term Differentiation} \\ &= 5x^4 - 6 \cdot \frac{d}{dx}x^7 && \text{Power Rule and Constant Multiple Rule} \\ &= 5x^4 - 6 \cdot (7x^6) && \text{Power Rule} \\ &= 5x^4 - 42x^6\end{aligned}$$

The last two examples illustrate that polynomials can be differentiated term by term.

Derivative of Polynomial Let $y = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$ be a polynomial. Then we have

$$\frac{dy}{dx} = na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \cdots + 2a_2 x + a_1.$$

$$\begin{aligned} \text{Proof } \frac{dy}{dx} &= \frac{d}{dx} (a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0) \\ &= \frac{d}{dx} a_n x^n + \frac{d}{dx} a_{n-1} x^{n-1} + \cdots + \frac{d}{dx} a_2 x^2 + \frac{d}{dx} a_1 x + \frac{d}{dx} a_0 && \text{Term by Term Differentiation} \\ &= a_n \frac{d}{dx} x^n + a_{n-1} \frac{d}{dx} x^{n-1} + \cdots + a_2 \frac{d}{dx} x^2 + a_1 \frac{d}{dx} x + 0 && \text{Constant Multiple Rule and} \\ &&& \text{Derivative of Constant} \\ &= a_n n x^{n-1} + a_{n-1} (n-1) x^{n-2} + \cdots + a_2 \cdot (2x) + a_1 \cdot 1 && \text{Power Rule} \quad \square \end{aligned}$$

Example Let $f(x) = 2x(x^2 - 5x + 7)$. Find the derivative of f at 2.

Explanation The question is to find $f'(2)$. We can find $f'(x)$ using one of the following two ways:

- product rule (will be discussed later);
- expand the expression to get a polynomial.

Then putting $x = 2$, we get the answer. Below we use the second method to find $f'(x)$.

$$\begin{aligned} \text{Solution } f'(x) &= \frac{d}{dx} (2x(x^2 - 5x + 7)) \\ &= \frac{d}{dx} (2x^3 - 10x^2 + 14x) \\ &= 2 \cdot (3x^2) - 10 \cdot (2x) + 14 \cdot 1 && \text{Derivative of Polynomial} \\ &= 6x^2 - 20x + 14 \end{aligned}$$

$$\begin{aligned} \text{The derivative of } f \text{ at } 2 \text{ is } f'(2) &= 6(2^2) - 20(2) + 14 \\ &= -2 \end{aligned} \quad \square$$

Product Rule Let f and g be functions with the same domain. Suppose that f and g are differentiable. Then the function fg is also differentiable. Moreover, we have

$$\frac{d}{dx}(fg)(x) = g(x) \cdot \frac{d}{dx} f(x) + f(x) \cdot \frac{d}{dx} g(x).$$

Explanation The function fg is defined by $(fg)(x) = f(x) \cdot g(x)$ for x belonging to the common domain A of f and g . The result means that if $f'(x)$ and $g'(x)$ exist for all $x \in A$, then $(fg)'(x) = g(x)f'(x) + f(x)g'(x)$ for all $x \in A$, that is, $(fg)' = gf' + fg'$.

Proof By definition, we have

$$\begin{aligned} (fg)'(x) &= \lim_{h \rightarrow 0} \frac{(fg)(x+h) - (fg)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} && \text{Definition of } fg \end{aligned}$$

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To find the limit, we use the following technique: “*subtract and add*” $f(x+h)g(x)$ in the numerator.

$$\begin{aligned}
 (fg)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h)g(x+h) - f(x+h)g(x)}{h} + \frac{f(x+h)g(x) - f(x)g(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(f(x+h) \cdot \frac{g(x+h) - g(x)}{h} \right) + \lim_{h \rightarrow 0} \left(g(x) \cdot \frac{f(x+h) - f(x)}{h} \right) && \text{Limit Rule (La4)} \\
 &= \lim_{h \rightarrow 0} f(x+h) \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} + \lim_{h \rightarrow 0} g(x) \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} && \text{Limit Rule (La5)} \\
 &= f(x)g'(x) + g(x)f'(x) && \text{Theorem 4.1.1 \& Limit Rule (La1)}
 \end{aligned}$$

In the last step, the first limit is found by substitution because f is continuous; the third limit is $g(x)$ because considered as a function of h , $g(x)$ is a constant.

Example Let $y = (x+1)(x^2+3)$. Find $\frac{dy}{dx}$.

Explanation The expression defining y is a product of two functions. So we can apply the product rule. Alternatively, we can expand the expression to get a polynomial and then differentiate term by term.

$$\begin{aligned}
 \text{Solution 1 } \frac{dy}{dx} &= \frac{d}{dx}((x+1)(x^2+3)) \\
 &= (x^2+3) \cdot \frac{d}{dx}(x+1) + (x+1) \cdot \frac{d}{dx}(x^2+3) && \text{Product Rule} \\
 &= (x^2+3) \cdot (1+0) + (x+1) \cdot (2x+0) && \text{Derivative of Polynomial} \\
 &= 3x^2 + 2x + 3
 \end{aligned}$$

$$\begin{aligned}
 \text{Solution 2 } \frac{dy}{dx} &= \frac{d}{dx}((x+1)(x^2+3)) \\
 &= \frac{d}{dx}(x^3 + x^2 + 3x + 3) \\
 &= 3x^2 + 2x + 3 && \text{Derivative of Polynomial}
 \end{aligned}$$

Quotient Rule Let f and g be functions with the same domain. Suppose that f and g are differentiable and that g has no zero in its domain. Then the function $\frac{f}{g}$ is also differentiable. Moreover, we have,

$$\frac{d}{dx} \left(\frac{f}{g} \right) (x) = \frac{g(x) \cdot \frac{d}{dx} f(x) - f(x) \cdot \frac{d}{dx} g(x)}{(g(x))^2}.$$

Explanation The condition “ g has no zero in its domain” means that $g(x) \neq 0$ for all $x \in \text{dom}(g)$. The function $\frac{f}{g}$ is defined by $\left(\frac{f}{g} \right) (x) = \frac{f(x)}{g(x)}$ for $x \in A$, where A is the common domain of f and g . The result means that if $f'(x)$ and $g'(x)$ exist for all $x \in A$, then $\left(\frac{f}{g} \right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}$ for all $x \in A$, that is, $\left(\frac{f}{g} \right)' = \frac{gf' - fg'}{g^2}$.

Proof The proof is similar to that for the product rule.

Example Let $y = \frac{x^2 + 3x - 4}{2x + 1}$. Find $\frac{dy}{dx}$.

$$\begin{aligned}
 \text{Solution } \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{x^2 + 3x - 4}{2x + 1} \right) \\
 &= \frac{(2x + 1) \cdot \frac{d}{dx}(x^2 + 3x - 4) - (x^2 + 3x - 4) \cdot \frac{d}{dx}(2x + 1)}{(2x + 1)^2} && \text{Quotient Rule} \\
 &= \frac{(2x + 1)(2x + 3) - (x^2 + 3x - 4)(2)}{(2x + 1)^2} && \text{Derivative of Polynomial} \\
 &= \frac{(4x^2 + 8x + 3) - (2x^2 + 6x - 8)}{(2x + 1)^2} \\
 &= \frac{2x^2 + 2x + 11}{(2x + 1)^2}
 \end{aligned}$$

Power Rule for Differentiation (negative integer version) Let n be a negative integer. Then the power function x^n is differentiable on $\mathbb{R} \setminus \{0\}$ and we have

$$\frac{d}{dx} x^n = nx^{n-1}.$$

Explanation Since n is a negative integer, it can be written as $-m$ where m is a positive integer. The function $x^n = x^{-m} = \frac{1}{x^m}$ is defined for all $x \neq 0$, that is, the domain of x^n is $\mathbb{R} \setminus \{0\}$.

Proof Let $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ be the function given by $f(x) = x^{-m}$, where $m = -n$. By definition, we have

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} \frac{1}{x^m} \\
 &= \frac{x^m \cdot \frac{d}{dx} 1 - 1 \cdot \frac{d}{dx} x^m}{(x^m)^2} && \text{Quotient Rule} \\
 &= \frac{x^m \cdot 0 - 1 \cdot mx^{m-1}}{x^{2m}} && \text{Derivative of Constant \& Power Rule (positive integer version)} \\
 &= -mx^{(m-1)-2m} \\
 &= -mx^{-m-1} \\
 &= nx^{n-1}
 \end{aligned}$$

Example Find an equation for the tangent line to the curve $y = \frac{3x^2 - 1}{x}$ at the point $(1, 2)$.

Explanation The curve is given by $y = f(x)$ where $f(x) = \frac{3x^2 - 1}{x}$. Since $f(1) = 2$, the point $(1, 2)$ lies on the curve. To find an equation for the tangent line, we have to find the slope at the point (and then use point-slope form). The slope at the point $(1, 2)$ is $f'(1)$. We can use rules for differentiation to find $f'(x)$ and then substitute $x = 1$ to get $f'(1)$.

Solution To find $\frac{dy}{dx}$, we can use quotient rule or term by term differentiation.

4.2. Rules for Differentiation

$$\begin{aligned}
 (\text{Method 1}) \quad \frac{dy}{dx} &= \frac{d}{dx} \frac{3x^2 - 1}{x} \\
 &= \frac{x \cdot \frac{d}{dx}(3x^2 - 1) - (3x^2 - 1) \cdot \frac{d}{dx}x}{x^2} && \text{Quotient Rule} \\
 &= \frac{x \cdot 3 \cdot (2x) - (3x^2 - 1) \cdot 1}{x^2} && \text{Derivative of Polynomial} \\
 &= \frac{3x^2 + 1}{x^2}
 \end{aligned}$$

$$\begin{aligned}
 (\text{Method 2}) \quad \frac{dy}{dx} &= \frac{d}{dx} ((3x^2 - 1)x^{-1}) \\
 &= \frac{d}{dx}(3x - x^{-1}) \\
 &= \frac{d}{dx}3x - \frac{d}{dx}x^{-1} && \text{Term by Term Differentiation} \\
 &= 3 - (-1)x^{-2} && \text{Derivative of Polynomial \& Power Rule} \\
 &= 3 + \frac{1}{x^2}.
 \end{aligned}$$

The slope of the tangent line at the point $(1, 2)$ is $\left. \frac{dy}{dx} \right|_{x=1} = 3 + \frac{1}{1} = 4$.

Equation for the tangent line: $y - 2 = 4(x - 1)$ (point-slope form)

$$4x - y - 2 = 0 \quad (\text{general linear form})$$

The next result is a special case of the general power rule. Since the square root function appears often in applied problems and the proof is not difficult, we give the result here.

Derivative of the Square Root Function The derivative of the square root function $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$, that is,

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}.$$

Explanation The domain of the square root function is $[0, \infty)$. Since the function is undefined on the left-side of 0, we can only consider differentiability of the function on $(0, \infty)$. The result means that $f'(x) = \frac{1}{2\sqrt{x}}$ for all $x \in (0, \infty)$ where $f(x) = \sqrt{x}$.

Proof Let $f : [0, \infty) \rightarrow \mathbb{R}$ be the function given by $f(x) = \sqrt{x}$. For every $x > 0$, note that if h is a small enough real number, then $x + h \in \text{dom}(f)$ and hence if h is a small enough non-zero real number, we have

$$\begin{aligned}
 \frac{f(x+h) - f(x)}{h} &= \frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} \\
 &= \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \\
 &= \frac{1}{\sqrt{x+h} + \sqrt{x}}
 \end{aligned}$$

which, by definition, implies that

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\
 &= \frac{1}{\lim_{h \rightarrow 0} (\sqrt{x+h} + \sqrt{x})} && \text{Limit Rule (La6)} \\
 &= \frac{1}{\lim_{h \rightarrow 0} \sqrt{x+h} + \lim_{h \rightarrow 0} \sqrt{x}} && \text{Limit Rule (La4)} \\
 &= \frac{1}{\sqrt{x+0} + \sqrt{x}} && \text{By continuity \& Limit Rule (La1)} \\
 &= \frac{1}{2\sqrt{x}}
 \end{aligned}$$

In the second last step, the first limit is found by substitution because the square root function is continuous; the second limit is the limit of a constant. $\square$

Example Let $y = \sqrt{x}(x+1)$. Find $\frac{dy}{dx}$.

$$\begin{aligned}
 \text{Solution } \frac{dy}{dx} &= \frac{d}{dx} \sqrt{x}(x+1) \\
 &= (x+1) \cdot \frac{d}{dx} \sqrt{x} + \sqrt{x} \cdot \frac{d}{dx} (x+1) && \text{Product Rule} \\
 &= (x+1) \cdot \frac{1}{2\sqrt{x}} + \sqrt{x} \cdot 1 && \text{Derivative of Square Root Function \& Derivative of Polynomial} \\
 &= \frac{\sqrt{x}}{2} + \frac{1}{2\sqrt{x}} + \sqrt{x} \\
 &= \frac{3\sqrt{x}}{2} + \frac{1}{2\sqrt{x}}
 \end{aligned}$$

Remark If we expand the expression defining y , we get $x^{\frac{3}{2}} + x^{\frac{1}{2}}$. The derivative can be found if we know the derivative of $x^{\frac{3}{2}}$.

Power Rule for Differentiation ($n + \frac{1}{2}$ version) Let n be an integer. Then the function $x^{n+\frac{1}{2}}$ is differentiable on $(0, \infty)$ and we have

$$\frac{d}{dx} x^{n+\frac{1}{2}} = \left(n + \frac{1}{2}\right) x^{n-\frac{1}{2}}.$$

Explanation Denoting the function by f , if $n = 0$, then f is the square root function; if n is positive, then the domain of f is $[0, \infty)$; if n is negative, then the domain of f is $(0, \infty)$. Putting $r = n + \frac{1}{2}$, then we have $f(x) = x^r$ and the result means that $f'(x) = rx^{r-1}$ for all $x > 0$. This result has the same form as the power rule and it will be referred to as the *power rule*.

Remark There is a more general result called the *General Power Rule* (see Chapter 9).

Proof Let f be the function given by $f(x) = x^{n+\frac{1}{2}}$. Note that $f(x) = x^n \cdot \sqrt{x}$ and so by the Product Rule and the

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Rule for Derivative of the Square Root Function, the function f is differentiable on $(0, \infty)$ and we have

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} (x^n \cdot \sqrt{x}) \\
 &= \sqrt{x} \cdot \frac{d}{dx} x^n + x^n \cdot \frac{d}{dx} \sqrt{x} && \text{Product Rule} \\
 &= \sqrt{x} \cdot nx^{n-1} + x^n \cdot \frac{1}{2\sqrt{x}} && \text{Power Rule \& Derivative Square Root Function} \\
 &= nx^{n-\frac{1}{2}} + \frac{1}{2}x^{n-\frac{1}{2}} \\
 &= \left(n + \frac{1}{2}\right)x^{n-\frac{1}{2}}
 \end{aligned}$$

Below, we redo the preceding example using the $n + \frac{1}{2}$ version of the power rule.

Example Let $y = \sqrt{x}(x + 1)$. Find $\frac{dy}{dx}$.

$$\begin{aligned}
 \text{Solution } \frac{dy}{dx} &= \frac{d}{dx} x^{\frac{1}{2}}(x + 1) \\
 &= \frac{d}{dx} (x^{\frac{3}{2}} + x^{\frac{1}{2}}) \\
 &= \frac{d}{dx} x^{\frac{3}{2}} + \frac{d}{dx} x^{\frac{1}{2}} && \text{Term by Term Differentiation} \\
 &= \frac{3}{2}x^{\frac{3}{2}-1} + \frac{1}{2}x^{\frac{1}{2}-1} && \text{Power Rule} \\
 &= \frac{3}{2}x^{\frac{1}{2}} + \frac{1}{2}x^{-\frac{1}{2}}
 \end{aligned}$$

Example Let $y = \frac{2x^2 - 3}{\sqrt{x}}$. Find $\frac{dy}{dx}$.

$$\begin{aligned}
 \text{Solution 1 } \frac{dy}{dx} &= \frac{d}{dx} \frac{2x^2 - 3}{\sqrt{x}} \\
 &= \frac{\sqrt{x} \cdot \frac{d}{dx} (2x^2 - 3) - (2x^2 - 3) \cdot \frac{d}{dx} \sqrt{x}}{(\sqrt{x})^2} && \text{Quotient Rule} \\
 &= \frac{\sqrt{x} \cdot 2 \cdot (2x) - (2x^2 - 3) \cdot \frac{1}{2\sqrt{x}}}{x} && \text{Derivative of Polynomial \& Derivative of Square Root Function} \\
 &= \frac{4x\sqrt{x} - x\sqrt{x} + \frac{3}{2\sqrt{x}}}{x} \\
 &= 3\sqrt{x} + \frac{3}{2x\sqrt{x}}
 \end{aligned}$$

$$\begin{aligned}
\text{Solution 2 } \frac{dy}{dx} &= \frac{d}{dx}(2x^2 - 3)x^{-\frac{1}{2}} \\
&= \frac{d}{dx}\left(2x^{\frac{3}{2}} - 3x^{-\frac{1}{2}}\right) \\
&= 2 \cdot \frac{d}{dx}x^{\frac{3}{2}} - 3 \cdot \frac{d}{dx}x^{-\frac{1}{2}} && \text{Term by Term Differentiation} \\
&= 2 \cdot \left(\frac{3}{2}x^{\frac{1}{2}}\right) - 3 \cdot \left(-\frac{1}{2}x^{-\frac{3}{2}}\right) && \text{Power Rule} \\
&= 3x^{\frac{1}{2}} + \frac{3}{2}x^{-\frac{3}{2}}
\end{aligned}$$

We close this section with the following “*Caution*” and “*Question*”. The “*caution*” points out a common mistake that many students made.

Caution $\frac{d}{dx}2^x \neq x \cdot 2^{x-1}$, we can’t apply the Power Rule because 2^x is not a power function; it is an exponential function (see Chapter 8).

Question Using rules discussed in this chapter, we can differentiate polynomial functions as well as rational functions. For example, to differentiate $f(x) = (x^2 + 5)^3$, we can first expand the cube to get a polynomial of degree 6 and then differentiate term by term. How about the following

- (1) $f(x) = (x^2 + 5)^{30}$?
- (2) $f(x) = \sqrt{x^2 + 5}$?

For (1), we can expand and then differentiate term by term (if we have enough patience). However, this doesn’t work for (2). Note that both functions are in the form $x \mapsto (x^2 + 5)^r$ which can be considered as composition of two functions:

$$x \mapsto (x^2 + 5) \mapsto (x^2 + 5)^r.$$

In Chapter 9, we will discuss the *chain rule*, a tool to handle this kind of differentiation.

Exercise 4.2

1. For each of the following y , find $\frac{dy}{dx}$.

- | | |
|----------------------------|----------------------------------|
| (a) $y = -\pi$ | (b) $y = 2x^9 + 3x$ |
| (c) $y = x^2 + 5x - 7$ | (d) $y = x(x - 1)$ |
| (e) $y = (2x - 3)(5 - 6x)$ | (f) $y = (x^2 + 5)^3$ |
| (g) $y = \frac{23}{x^4}$ | (h) $y = \frac{x-1}{x}$ |
| (i) $y = \frac{x-1}{x+1}$ | (j) $y = \sqrt{x}(\sqrt{x} + 1)$ |

2. For each of the following f , find $f'(a)$ for the given a .

- | | |
|--|---|
| (a) $f(x) = x^3 - 4x, \quad a = 1$ | (b) $f(x) = \frac{2}{x^3} + \frac{4}{x}, \quad a = 2$ |
| (c) $f(x) = 3x^{\frac{1}{3}} - 5x^{\frac{2}{3}}, \quad a = 27$ | (d) $f(x) = \pi x^2 - 2\sqrt{x}, \quad a = 4$ |
| (e) $f(x) = (x^2 + 3)(x^3 + 2), \quad a = 1$ | (f) $f(x) = \frac{x^2 + 1}{2x - 3}, \quad a = 2$ |
| (g) $f(x) = \frac{x^2 + 3x - 5}{x^2 - 7x + 5}, \quad a = 1$ | |

4.3. Higher-Order Derivatives

3. Consider the curve given by $y = 3x^4 - 6x^2 + 2$.
- Find the slope of the curve at the point whose x -coordinate is 2.
 - Find an equation for the tangent line at the point $(1, -1)$.
 - Find the point(s) on the curve at which the tangent line is horizontal.
4. It is known that $\frac{d}{dx} \sin x = \cos x$. For each of the following y , find $\frac{dy}{dx}$.
- $y = x \sin x$
 - $y = \frac{\sin x}{x+1}$
 - $y = \frac{x^2+1}{\sin x}$
 - $y = x(x+2) \sin x$
5. Let f be a differentiable function.
- Use product rule to show that $\frac{d}{dx} f(x)^2 = 2f(x) \cdot \frac{d}{dx} f(x)$.
 - Use product rule and the result in (a) to show that $\frac{d}{dx} f(x)^3 = 3f(x)^2 \cdot \frac{d}{dx} f(x)$.
 - Can you guess (and prove) a formula for $\frac{d}{dx} f(x)^n$, where n is a positive integer?
6. (a) Use the result in 5(a) to find $\frac{d}{dx}(x^3 + 5x^2 - 2)^2$.
- (b) Use the result in 5(b) to find $\frac{d}{dx}(x^2 + 5)^3$. Compare your answer with that in Q.1(f).

4.3 Higher-Order Derivatives

Let f be a function that is differentiable at some points belonging to $\text{dom}(f)$. Then f' is a function.

- If, in addition, f' is differentiable at some points belonging to $\text{dom}(f')$, then the derivative of f' exists and is denoted by f'' ; it is the function given by $f''(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h}$ and is called the *second derivative* of f .
- If, in addition, f'' is differentiable at some points belonging to $\text{dom}(f'')$, then the derivative of f'' is denoted by f''' , called the *third derivative* of f .
- In general, the n -th derivative of f (where n is a positive integer), denoted by $f^{(n)}$, is defined to be the derivative of the $(n-1)$ -th derivative of f (where the 0-th derivative of f means f). For $n = 1$, the first derivative of f is simply the derivative f' of f . For $n > 1$, $f^{(n)}$ is called a *higher-order derivative* of f .

Notation Similar to first order derivative, we have different notations for second order derivative of f .

$$y'', \quad f'', \quad \frac{d^2y}{dx^2}, \quad D^2f, \quad D^2y, \quad f''(x) \quad \text{and} \quad \frac{d^2}{dx^2}f(x).$$

Readers may compare these with that on page 109. Similarly, we also have different notations for other higher-order derivatives.

Example Let $f(x) = 5x^3 - 2x^2 + 6x + 1$. Find the derivative and all the higher-order derivatives of f .

Explanation The question is to find for each positive integer n , the domain of the n -th derivative of f and a formula for $f^{(n)}(x)$. To find $f'(x)$, we can apply differentiation term by term. To find $f''(x)$, by definition, we have $f''(x) = \frac{d}{dx}f'(x)$ which can be simplified using the result for $f'(x)$ and rules for differentiation.

$$\begin{aligned}
\text{Solution} \quad f'(x) &= \frac{d}{dx}(5x^3 - 2x^2 + 6x + 1) \\
&= 15x^2 - 4x + 6 && \text{Derivative of Polynomial} \\
f''(x) &= \frac{d}{dx}(15x^2 - 4x + 6) \\
&= 30x - 4 && \text{Derivative of Polynomial} \\
f'''(x) &= \frac{d}{dx}(30x - 4) \\
&= 30 && \text{Derivative of Polynomial} \\
f^{(4)}(x) &= 0 && \text{Derivative of Constant}
\end{aligned}$$

From this we see that for $n \geq 4$, $f^{(n)}(x) = 0$. Moreover, for every positive integer n , the domain of $f^{(n)}$ is $\mathbb{R}$. $\square$

Example Let $f(x) = \frac{x^3 - 1}{x}$. Find $f'(3)$ and $f''(-4)$.

Explanation

- To find $f'(3)$, we find $f'(x)$ first and then substitute $x = 3$. Although $f(x)$ is written as a quotient of two functions, it is better to find $f'(x)$ by expanding $(x^3 - 1)x^{-1}$.
- To find $f''(-4)$, we find $f''(x)$ first and then substitute $x = -4$. To find $f''(x)$, we differentiate the result obtained for $f'(x)$.

$$\begin{aligned}
\text{Solution} \quad f'(x) &= \frac{d}{dx}((x^3 - 1)x^{-1}) \\
&= \frac{d}{dx}(x^2 - x^{-1}) \\
&= 2x - (-1)x^{-2} && \text{Term by Term Differentiation \& Power Rule} \\
&= 2x + x^{-2} \\
f'(3) &= 2 \cdot (3) + 3^{-2} \\
&= \frac{55}{9} \\
f''(x) &= \frac{d}{dx}(2x + x^{-2}) && \text{By result for } f'(x) \\
&= 2 + (-2)x^{-3} && \text{Term by Term Differentiation \& Power Rule} \\
&= 2 - 2x^{-3} \\
f''(-4) &= 2 - 2 \cdot (-4)^{-3} \\
&= \frac{65}{32}
\end{aligned}$$

Meaning of Second Derivative

- The graph of $y = f(x)$ is a curve. Note that $f'(x) = \frac{dy}{dx}$ is the slope function; it is the rate of change of y with respect to x . Since $f''(x) = \frac{d^2y}{dx^2}$ is the derivative of the slope function, it is the rate of change

4.3. Higher-Order Derivatives

of slope and is related to a concept called *convexity* (*bending*) of a curve. More details can be found in Chapter 5.

- If $x = t$ is time and if $y = s(t)$ is the displacement function of a moving object, then $s'(t) = \frac{ds}{dt}$ is the velocity function. The derivative of velocity is $s''(t)$ or $\frac{d^2s}{dt^2}$; it is the rate of change of the velocity (function), that is, the *acceleration* (function).

Exercise 4.3

1. For each of the following y , find $\frac{d^2y}{dx^2}$.
 - (a) $y = x^3 - 3x^2 + 4x - 1$
 - (b) $y = (2x + 3)(4 - x)$
 - (c) $y = \sqrt{x}(1 + x)$
 - (d) $y = \frac{1 - 2x}{x^2}$
 - (e) $y = (x^3 + 1)^2$
2. For each of the following f , find $f''(a)$ for the given a .
 - (a) $f(x) = 7x^6 - 8x^5 + 15x$, $a = 1$
 - (b) $f(x) = x^2(1 - 2x)$, $a = 2$
 - (c) $f(x) = (2 + 3x)^2$, $a = 0$
3. Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial of degree n .
 - (a) Find $f(0)$ and $f'(0)$.
 - (b) Find $f^{(n)}(x)$ and $f^{(n+1)}(x)$.